Investigating Word Associations in Historical Mental Health Discussions Using MentalBERT

Abdelkarim Zirar · MA Data and Discourse Studies · February 12, 2025

Embedded Meaning: Word Embeddings in Theory and Practice

INTRODUCTION

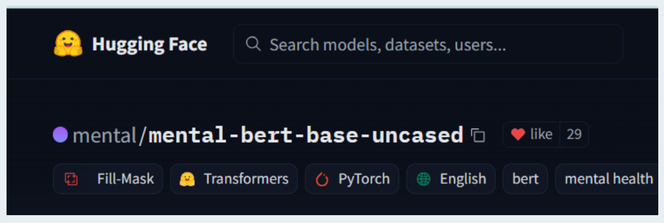
Exploring how a modern mental health-specific model interprets historical language can provide insights into word associations within past mental health discourse.

This study examines the semantic relationships between terms connected to mental health in historical literature using word embedding techniques. It also looks at how MentalBERT grasps this discourse and additionally gains valuable insights and perceptions. This approach allows us to uncover dominant themes and emotional connotations. The findings contribute to a better understanding of how historical language around mental health aligns with or diverges from current contemporary perspectives.

Datasets

This study applies MentalBERT, a transformer-based language model trained on Reddit posts pertaining to mental health. This model is utilized to analyze word embeddings in a historical corpus.

This study employs a dataset that consists of 188 digitized newspaper articles/pages linked to mental health topics and terminology. The chosen newspapers were published between 1858 and 1963 and are sourced from the Chronicling America Historic American Newspapers collection in The Library of Congress.



METHODS

Data Collection:

- The corpus includes discussions of mental health-related topics from various newspaper sources and headlines over time, providing an overview into historical perspectives on mental illness, treatment, and stigma.
- Through an advanced search, keywords were used to identify the newspapers.

Model Selection:

- The MentalBERT model was loaded using Huggingface's Transformers library. It was chosen because it is specifically trained on mental health-related language.

Preprocessing and Tokenization:

- Although it was a challenge due to OCR errors, the historical corpus was cleaned and tokenized to prepare it for embedding extraction.
- Tokenization was performed using the MentalBERT tokenizer, ensuring compatibility with the model's vocabulary and structure.

Cosine Similarity:

- The top 5 most semantically similar words were identified for each mental health term based on cosine similarity scores.

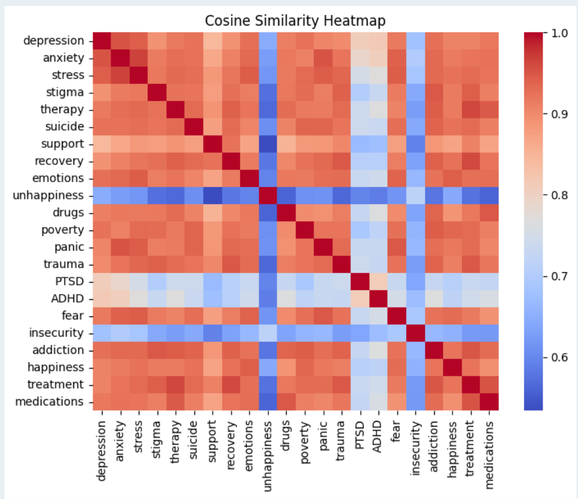
Visualization Techniques:

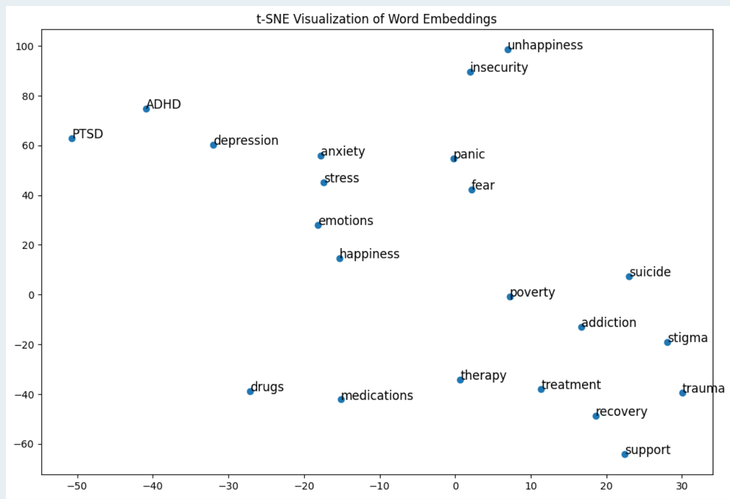
- **Cosine Similarity Heatmaps** generated using seaborn.heatmap().
- **t-SNE graphs** created with sklearn.manifold.TSNE.
- **Network Graphs** to illustrate relationships between key mental health terms and their top associations in the historical corpus. Constructed using networkx.

RESULTS

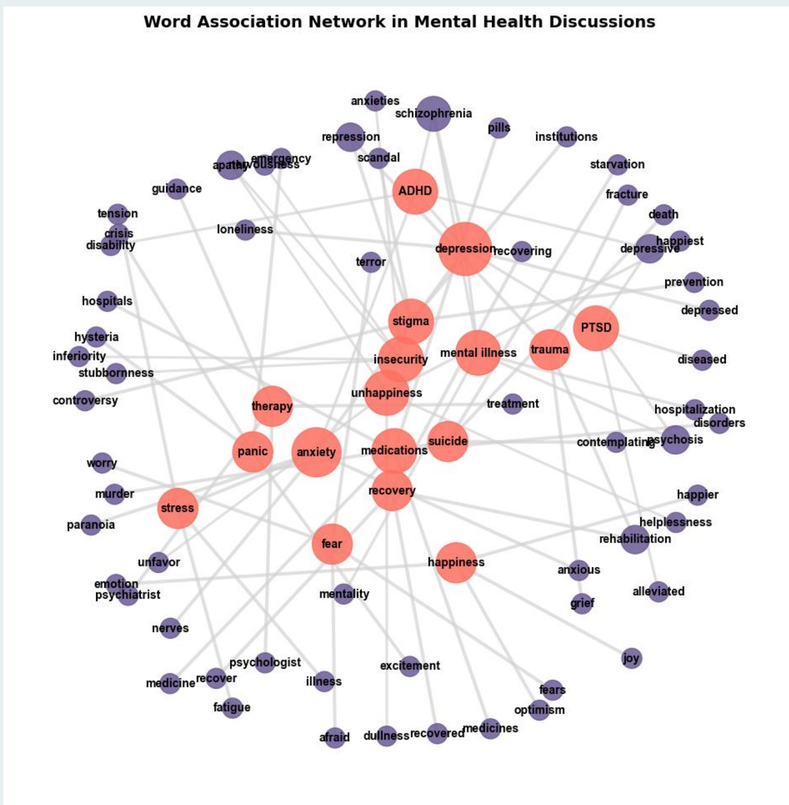
1. Understanding MentalBERT's Word Associations

This is useful because it shows how a mental-health specific model understands word relationships. The model was provided with a predefined set of key terms, including 'depression,' 'anxiety,' 'stress,' 'stigma,' 'therapy,' 'suicide,' 'support,' 'recovery,' 'emotions,' 'unhappiness,' 'drugs,' 'poverty,' 'panic,' 'trauma,' 'PTSD,' 'ADHD,' 'fear,' 'insecurity,' 'addiction,' 'happiness,' 'treatment,' and 'medications.'





2. Historical Mental Health Discourse



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word_associations = {
    "depression": [("anxiety", 0.9511), ("loneliness", 0.9480), ("schizophrenia", 0.9456), ("depressed", 0.9452)],
    "anxiety": [("stress", 0.9656), ("anxious", 0.9594), ("paranoia", 0.9533), ("nerves", 0.9519)],
    "stress": [("anxiety", 0.9656), ("fatigue", 0.9540), ("tension", 0.9533), ("illness", 0.9510)],
    "stigma": [("prevention", 0.9537), ("controversy", 0.9534), ("institutions", 0.9516), ("scandal", 0.9516), ("repression", 0.9507)],
    "therapy": [("treatment", 0.9621), ("psychologist", 0.9562), ("psychiatrist", 0.9538), ("guidance", 0.9523)],
    "suicide": [("death", 0.9480), ("starvation", 0.9442), ("murder", 0.9436), ("contemplating", 0.9433)],
    "recovery": [("recover", 0.9757), ("recovering", 0.9709), ("recovered", 0.9704), ("rehabilitation", 0.9644)],
    "unhappiness": [("happiest", 0.8206), ("helplessness", 0.7411), ("dullness", 0.7395), ("unfavor", 0.7351), ("apathy", 0.7301)],
    "panic": [("hysteria", 0.9564), ("crisis", 0.9546), ("excitement", 0.9545), ("emergency", 0.9543)],
    "trauma": [("grief", 0.9609), ("rehabilitation", 0.9571), ("fracture", 0.9563), ("repression", 0.9557)],
    "PTSD": [("depressive", 0.8261), ("diseased", 0.8248), ("psychosis", 0.8110), ("depression", 0.8086), ("alleviated", 0.8067)],
    "ADHD": [("depression", 0.8173), ("schizophrenia", 0.8163), ("anxiety", 0.8056), ("depressive", 0.7989), ("disability", 0.7906)],
    "fear": [("fears", 0.9654), ("terror", 0.9584), ("afraid", 0.9535), ("worry", 0.9531)],
    "insecurity": [("stubbornness", 0.7510), ("nervousness", 0.7451), ("inferiority", 0.7371), ("anxieties", 0.7341), ("apathy", 0.7326)],
    "happiness": [("joy", 0.9569), ("happier", 0.9523), ("emotion", 0.9496), ("optimism", 0.9489)],
    "medications": [("medicines", 0.9772), ("pills", 0.9623), ("disorders", 0.9622), ("hospitals", 0.9612), ("medicine", 0.9602)],
    "mental illness": [("hospitalization", 0.8279), ("psychosis", 0.8262), ("depression", 0.8255), ("schizophrenia", 0.8248), ("mentality", 0.8226)]
}
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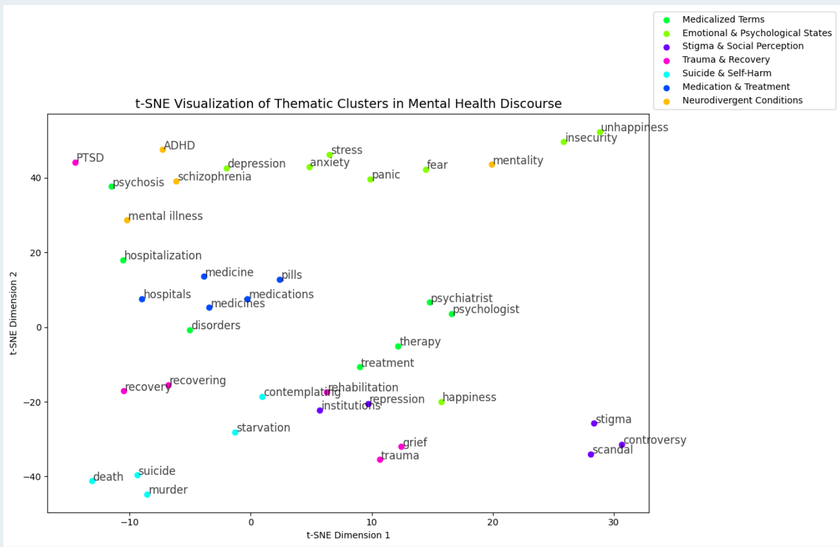

DISCUSSION

- The results highlight historical perceptions and terminology in mental health discussions.
- Terms like *PTSD* and *ADHD* show weaker semantic relationships, reflecting their limited recognition in historical discourse.
- The study demonstrates how MentalBERT can be used to interpret historical mental health language, providing a unique way to analyze historical perceptions using modern NLP techniques.
- MentalBERT effectively groups related mental health terms based on context, as seen in high cosine similarities between words like *depression-anxiety* (0.9511) and *stress-anxiety* (0.9656), reflecting real-world co-occurrence in discussions.

- Some word associations (like PTSD being linked to psychosis and depressive) might not reflect how these terms were understood historically.
- Instead, these associations may reveal how the modern model—trained on contemporary discussions—interprets historical language.
- This suggests that MentalBERT's interpretation of historical discourse is shaped by present-day understandings of mental health, rather than how these terms were originally used in historical contexts.

REFERENCE

- Ji, S., Zhang, T., Ansari, L., Fu, J., Tiwari, P., & Cambria, E. (2022). MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare. *Proceedings of LREC*.
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